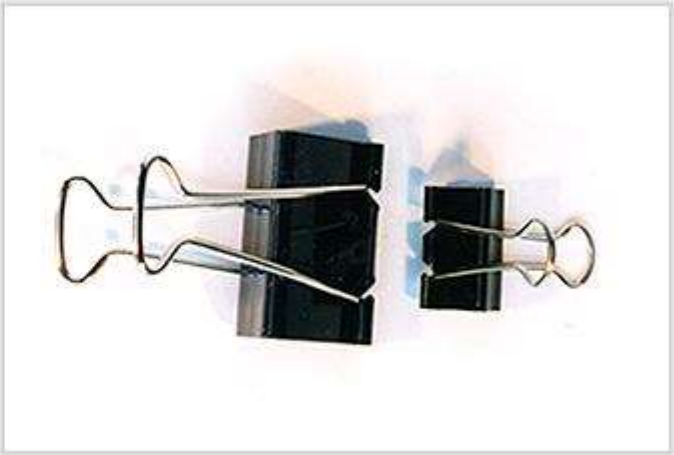
Description

Image





Caption

1. Bearing made of high carbon steel. © ANSYS, Inc. 2. Drawing board clips made of high carbon steel. © ANSYS, Inc.

The material

High carbon steels (0.7-1.7% carbon) harden when quenched - a quality that gives great control over properties. High carbon steels achieve hardness sufficient for them to be used as cutting tools, chisels and cables, and "piano wire" - the metal strings of pianos and violins.

Compositional summary

Fe/0.7 - 1.7%C

General properties

Density		7.8e3			kg/m^3
Price	*	0.911	-	1.02	GBP/kg
Date first used		1610			

Mechanical properties

Young's modulus		200	-	220	GPa
Shear modulus		77	-	85	GPa
Bulk modulus		160	-	180	GPa
Poisson's ratio		0.28	-	0.29	
Yield strength (elastic limit)		433	-	924	MPa
Tensile strength		721	-	1.39e3	MPa
Compressive strength	*	433	-	924	MPa
Elongation		9.6	-	21	% strain
Hardness - Vickers		220	-	419	HV
Fatigue strength at 10^7 cycles	*	341	-	539	MPa

Fracture toughness	*	33	-	75.6	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	3.7e-4	-	7.6e-4	

Thermal properties

Melting point		1.3e3	-	1.48e3	°C
Maximum service temperature	*	272	-	340	°C
Minimum service temperature	*	-48.2	-	-3.15	°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		47	-	53	W/m.°C
Specific heat capacity		448	-	510	J/kg.°C
Thermal expansion coefficient		11	-	13.2	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		17	-	20	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		2	-	3	
Formability		4	-	5	
Machinability		3	-	4	
Weldability		2	-	4	
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Acceptable			
Water (salt)		Limited use			
Soils, acidic (peat)		Acceptable			
Soils, alkaline (clay)		Acceptable			
Wine		Unacceptable			

Durability: acids

Acetic acid (10%)		Limited use			
Acetic acid (glacial)		Unacceptable			
Citric acid (10%)		Unacceptable			

Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Unacceptable
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Limited use

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Acceptable
Formaldehyde (40%)	Unacceptable
Glycerol	Excellent
Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Acceptable
Fluorine (gas)	Excellent
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Limited use
Rural atmosphere	Acceptable
Marine atmosphere	Limited use
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Acceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component		2.23e9		tonne/yr
Reserves, principal component		1.59e11		tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	*	2.24	-	2.46	kg/kg
Embodied energy (virgin grade production)	*	30.4	-	33.6	MJ/kg
Water usage	*	43.5	-	48.1	l/kg

Material processing: energy

Metal casting, embodied energy	*	47.2	-	54.7	MJ/kg
Metal extrusion, hot, embodied energy	*	17.6	-	23.5	MJ/kg
Metal rolling and forging, embodied energy	*	9.28	-	12.3	MJ/kg
Wire drawing, embodied energy	*	4.38	-	4.83	MJ/kg
Metal powder forming, embodied energy	*	62.4	-	83	MJ/kg
Chemical vapor deposition (CVD), embodied energy	*	1e4	-	1.13e4	MJ/kg
Machining, coarse, embodied energy	*	17.1	-	19.5	MJ/kg

Machining, fine, embodied energy	* 22.1	-	26.2	MJ/kg
Machining, grinding, embodied energy	* 27.1	-	33	MJ/kg
Machining, laser, CO2, embodied energy	* 1.19	-	1.34	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.39	-	3.96	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.15	-	1.54	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.607	-	0.806	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.286	-	0.317	kg/kg
Metal powder forming, climate change (CO2-eq)	* 5.09	-	6.77	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 823	-	926	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.12	-	1.28	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.45	-	1.72	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.78	-	2.16	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.098	-	0.11	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 0.605	-	0.664	kg/kg
Embodied energy, recycling	* 9.42	-	10.4	MJ/kg
Recycle fraction in current supply	40			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

The production energy of steel is comparatively low - per unit weight, about a half that of polymers; per unit volume, though, twice as much. Carbon steels are easy to recycle, and the energy to do so is small.

Supporting information

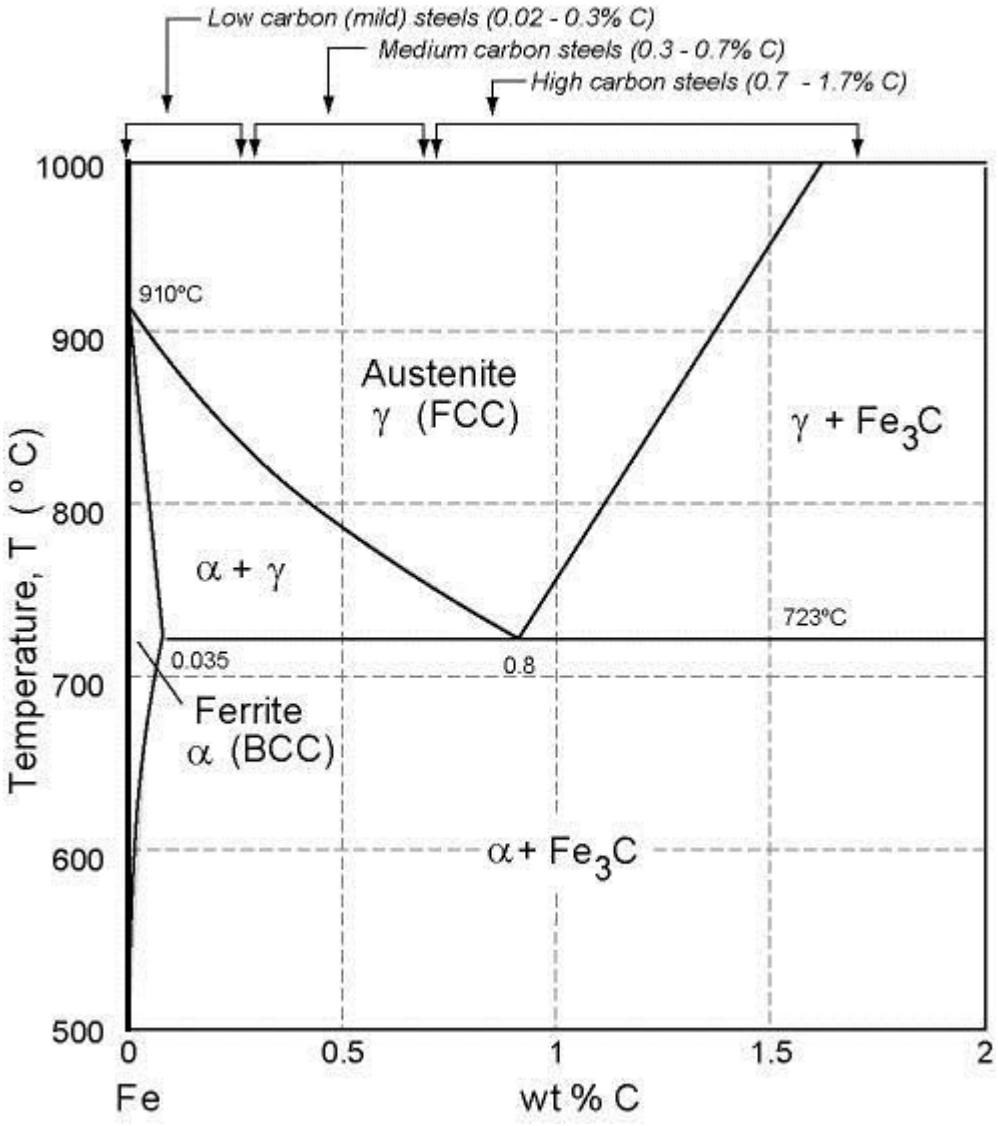
Design guidelines

Hardenability measures the degree to which it can be hardened in thick sections; plain carbon steels have poor hardenability - additional alloying elements are used to increase it (see Low alloy steels).

Technical notes

The two standard classifications for steels, the AISI and the SAE standards, have now been merged. In the SAE-AISI system, each steel has a four-digit code. The first two digits indicate the major alloying elements. The second two give the amount of carbon, in hundredths of a percent. Thus the plain carbon steels have designations starting 10xx, 11xx, 12xx or 14xxx, depending on how much manganese, sulfur and phosphorus they contain. The common low-carbon steels have the designations 1015,1020, 1022, 1117,1118; the common medium carbon steels are 1030,1040, 1050, 1060, 1137, 1141, 1144 and 1340; the common high alloy steels are 1080and 1095.

Phase diagram



Phase diagram description

High carbon steels are alloys of iron (Fe) with 0.7 - 1.7% carbon (C), for which this is the phase diagram.

Typical uses

Cutting tools, high performance bearings, cranks and shafts, springs, knives and scissors, rail track.

Links

ProcessUniverse

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Producers

Reference

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